

state purification dynamics.

8. M. Ippoliti, M. J. Gullans, S. Gopalakrishnan, D. A. Huse, and V. Khemani, “Entanglement Phase Transitions in Measurement-Only Dynamics,” *Phys. Rev. X* **11**, 011030 (2021). [BPJ] [ARXIV]

Summary. Measurement-only circuit numerics show that competing noncommuting projectors can generate both critical and highly entangled phases without unitary gates. This is a direct conceptual ancestor of measurement frustration and of controllers built from incompatible fermionic counting modes.

9. Y. Li, X. Chen, A. W. W. Ludwig, and M. P. A. Fisher, “Conformal Invariance and Quantum Nonlocality in Critical Hybrid Circuits,” *Phys. Rev. B* **104**, 104305 (2021). [BOTH] [ARXIV]

Summary. Numerical data demonstrate spacetime conformal covariance of entanglement observables at a hybrid-circuit critical point. The work provides stringent diagnostics for deciding whether apparent logarithmic scaling in monitored fermions is a true critical phase or a finite-scale crossover.

II. SYMMETRY, CONSERVATION LAWS, AND CHARGE SHARPENING

How conserved quantities and symmetry reorganize trajectory phases and observables.

10. Y. Bao, S. Choi, and E. Altman, “Symmetry Enriched Phases of Quantum Circuits,” *Ann. Phys.* **435**, 168618 (2021). [BPJ] [ARXIV]

Summary. Replica field theory and numerics show that monitored-circuit ensembles possess enlarged dynamical symmetries and phases with no equilibrium analogue. Its Gaussian-fermion example predicts a U(1) critical regime and Kosterlitz-Thouless transitions, making symmetry of the trajectory ensemble—not only microscopic symmetry—central.

11. F. Barratt, U. Agrawal, S. Gopalakrishnan, D. A. Huse, R. Vasseur, and A. C. Potter, “Field Theory of Charge Sharpening in Symmetric Monitored Quantum Circuits,” *Phys. Rev. Lett.* **129**, 120604 (2022). [BPJ] [ARXIV]

Summary. A controlled continuous-time replica field theory predicts a critical charge-fuzzy phase and a modified Kosterlitz-Thouless sharpening transition. Matrix-product-state numerics support the theory for discrete projective circuits, bridging continuous weak monitoring and circuit implementations.

12. U. Agrawal, A. Zabalo, K. Chen, J. H. Wilson, A. C. Potter, J. H. Pixley, S. Gopalakrishnan, and R. Vasseur, “Entanglement and Charge Sharpening Transitions in U(1) Symmetric Monitored Quantum Circuits,” *Phys. Rev. X* **12**, 041002 (2022). [BPJ] [ARXIV]

Summary. Exact numerics and a constrained-random-walk mapping identify a charge-sharpening transition within the volume-law phase of U(1)-symmetric monitored circuits. The work separates the learnability of a conserved charge from entanglement destruction and supplies scalable ancilla probes for both transitions.

13. S. Majidy, U. Agrawal, S. Gopalakrishnan, A. C. Potter, R. Vasseur, and N. Yunger Halpern, “Critical Phase and Spin Sharpening in SU(2)-Symmetric Monitored Quantum Circuits,” *Phys. Rev. B* **108**, 054307 (2023). [BPJ] [ARXIV]

Summary. Exact numerics and an effective statistical model find a critical phase with diffusive purification and a distinct spin-sharpening transition under SU(2)-symmetric monitoring. The result shows that non-Abelian conservation qualitatively changes even measurement-only dynamics.

14. H. Guo, M. S. Foster, C.-M. Jian, and A. W. W. Ludwig, “Field Theory of Monitored Interacting Fermion Dynamics with Charge Conservation,” *Phys. Rev. B* **112**, 064304 (2025). [BPJ] [ARXIV]

Summary. A replica Keldysh field theory shows that interactions reduce the large free-fermion replica symmetry to permutation and replica-resolved charge symmetries. The resulting theory contains both an entanglement transition and a Kosterlitz-Thouless charge-sharpening transition, explaining why the free and interacting one-dimensional problems differ sharply.

15. I. Poboiko, P. Poepperl, I. V. Gornyi, and A. D. Mirlin, “Measurement-Induced Transitions for Interacting Fermions,” *Phys. Rev. B* **111**, 024204 (2025). [BPJ] [ARXIV]

Summary. A replicated Keldysh nonlinear sigma model and matrix-product-state numerics show interaction-induced information-charge separation and stabilization of volume-law entanglement. The analysis predicts measurement transitions in any dimension for interacting fermions, unlike the short-range free-fermion chain.

III. MONITORED FREE FERMIONS, GAUSSIAN TRAJECTORIES, AND UNRAVELINGS

The trajectory-level phases, field theories, and dependence on the physical instrument.

16. X. Cao, A. Tilloy, and A. De Luca, “Entanglement in a Fermion Chain under Continuous Monitoring,” *SciPost Phys.* **7**, 024 (2019). [FGTN] [ARXIV]

Summary. Stochastic correlation-matrix simulations of a continuously monitored free-fermion chain found logarithmic entanglement over accessible scales and derived nonlinear evolution equations for two-point correlations. It established the computational and conceptual baseline for continuous Gaussian monitoring.

17. O. Alberton, M. Buchhold, and S. Diehl, “Entanglement Transition in a Monitored Free-Fermion Chain: From Extended Criticality to Area Law,” *Phys. Rev. Lett.* **126**, 170602 (2021). [BOTH] [ARXIV]

Summary. Trajectory simulations reported an extended critical regime and a transition to area-law entanglement as monitoring increases. Subsequent NLSM work argues that the weak-monitoring regime ultimately localizes at exponentially large scales, making this study a central reference for the interpretation of finite-size data.

18. B. Ladewig, S. Diehl, and M. Buchhold, “Monitored Open Fermion Dynamics: Exploring the Interplay of Measurement, Decoherence, and Free Hamiltonian Evolution,” *Phys. Rev. Research* **4**, 033001 (2022). [NEW] [ARXIV]

Summary. Analytical arguments and trajectory numerics distinguish information-preserving monitoring from unobserved decoherence in open free-fermion chains. The work is important for physical implementations because the same averaged Lindblad equation can hide sharply different conditional entanglement dynamics.

19. G. Piccitto, D. Rossini, and A. Russomanno, “The Impact of Different Unravelings in a Monitored System of Free Fermions,” *Eur. Phys. J. B* **97**, 90 (2024). [NEW] [ARXIV]

Summary. Correlation-matrix numerics compare quantum-jump and quantum-state-diffusion unravelings of the same dephasing Lindbladian. Both show a measurement-operator bifurcation and multifractal Slater components, but their distributions and trajectory transitions differ, demonstrating that conditional physics is not fixed by the master equation alone.

20. M. Fava, L. Piroli, D. Bernard, and A. Nahum, “Monitored Fermions with Conserved U(1) Charge,” *Phys. Rev. Research* **6**, 043246 (2024). [BPJ] [ARXIV]

Summary. Field theory and numerics analyze monitored fermions with exact U(1) conservation, emphasizing the coupling between charge diffusion and replica degrees of freedom. The work sharpens which features are universal to number-conserving Gaussian trajectories and which require interacting replica sectors.

21. C. Y. Leung, D. Meidan, and A. Romito, “Theory of Free Fermions Dynamics under Partial Postselected Monitoring,” *Phys. Rev. X* **15**, 021020 (2025). [NEW] [ARXIV]

Summary. A microscopic stochastic Schrodinger equation continuously interpolates between fully stochastic monitoring and postselected non-Hermitian evolution for measured Majorana parities. Bosonization, renormalization, and numerics find that non-Hermitian criticality tolerates weak stochasticity but crosses abruptly to the monitored universality class at finite partial postselection.

22. C. Niederegger, T. Vovk, E. Starchl, and L. M. Sieberer, “Absence of Measurement- and Unraveling-Induced Entanglement Transitions in Continuously Monitored One-Dimensional Free Fermions,” *Phys. Rev. B* **113**, 144317 (2026). [NEW] [ARXIV]

Summary. Replica Keldysh theory and trajectory numerics vary an unraveling phase continuously between number measurement and unitary noise. For every genuinely informative unraveling the asymptotic state is area-law entangled beyond an exponentially large scale, establishing that numerically visible critical behavior is a crossover rather than a finite phase in this model.

IV. GAUSSIAN CIRCUITS, TENSOR NETWORKS, LOCALIZATION, AND MULTIFRACTALITY

The transfer-matrix and disordered-fermion machinery underlying monitored Gaussian dynamics.

23. X. Chen, Y. Li, M. P. A. Fisher, and A. Lucas, “Emergent Conformal Symmetry in Nonunitary Random Dynamics of Free Fermions,” *Phys. Rev. Research* **2**, 033017 (2020). [FGTN] [ARXIV]

Summary. Gaussian-circuit numerics found logarithmic entanglement and spacetime conformal scaling, while a nonlinear master equation explains two-point dynamics. Later field theories reinterpret much of this regime as an exceptionally long crossover, so this paper is essential for understanding the live finite-size controversy.

24. M. Buchhold, Y. Minoguchi, A. Altland, and S. Diehl, “Effective Theory for the Measurement-Induced Phase Transition of Dirac Fermions,” *Phys. Rev. X* **11**, 041004 (2021). [NEW] [ARXIV]

Summary. Replica field theory treats a continuously monitored Dirac chain as a transition in the dark state of an effective non-Hermitian theory. It inaugurated the Buchhold-Diehl continuum approach and clarifies which apparent transitions depend on long-range

structure or additional tuning.

- 25.** C.-M. Jian, B. Bauer, A. Keselman, and A. W. W. Ludwig, “Criticality and Entanglement in Nonunitary Quantum Circuits and Tensor Networks of Noninteracting Fermions,” *Phys. Rev. B* **106**, 134206 (2022). [BOTH] [ARXIV]

Summary. The paper proves a correspondence between nonunitary Gaussian circuits in d dimensions, Gaussian tensor networks in $d+1$ dimensions, and static disordered free fermions in the ten Altland-Zirnbauer classes. Numerics and field theory identify boundary entanglement criticality with Anderson localization and metallic phases, providing the main conceptual seed for the present literature map.

- 26.** C.-M. Jian, H. Shapourian, B. Bauer, and A. W. W. Ludwig, “Measurement-Induced Entanglement Transitions in Quantum Circuits of Noninteracting Fermions: Born-Rule versus Forced Measurements,” arXiv:2302.09094 (2023), preprint. [BPJ] [ARXIV]

Summary. Numerics and symmetry analysis show that Born-rule and forced Gaussian measurements in class DIII have different critical exponents despite sharing the same Altland-Zirnbauer class. The field-theory explanation is the distinct replica limits n to 1 and n to 0, separating monitored universality from ordinary Anderson localization.

- 27.** I. Poboiko, P. Poepperl, I. V. Gornyi, and A. D. Mirlin, “Theory of Free Fermions under Random Projective Measurements,” *Phys. Rev. X* **13**, 041046 (2023). [BPJ] [ARXIV]

Summary. Replica Keldysh theory derives an $SU(R)$ nonlinear sigma model for local projective number measurements and connects entanglement to particle-number cumulants. Renormalization predicts ultimate area-law localization in one dimension at any nonzero measurement rate, with exponentially large crossover lengths at weak monitoring.

- 28.** J. Merritt and L. Fidkowski, “Entanglement Transitions with Free Fermions,” *Phys. Rev. B* **107**, 064303 (2023). [BPJ] [ARXIV]

Summary. Gaussian-state numerics generalize the completely packed loop model with crossings beyond fine-tuned Majorana swaps and show that its Goldstone and area-law phases persist. Notably, a Goldstone phase survives at finite rates even for commuting measurements once generic unitary evolution is present.

- 29.** K. Klocke and M. Buchhold, “Majorana Loop Models for Measurement-Only Quantum Circuits,” *Phys. Rev. X* **13**, 041028 (2023). [BPJ] [ARXIV]

Summary. Loop-model mappings and large-scale numerics classify measurement-only Majorana and Clifford circuits by world-line orientability. The analysis connects microscopic measurement choices to percolation, Ising, BKT, and CPLC universality classes, including the characteristic log-squared entanglement Goldstone phase.

- 30.** M. Fava, L. Piroli, T. Swann, D. Bernard, and A. Nahum, “Nonlinear Sigma Models for Monitored Dynamics of Free Fermions,” *Phys. Rev. X* **13**, 041045 (2023). [BPJ] [ARXIV]

Summary. A systematic replica construction derives nonlinear sigma models for several monitored free-fermion symmetry classes and explains Goldstone phases and localization transitions. It is one of the two foundational 2023 field theories connecting monitored Gaussian dynamics to disordered metals.

- 31.** I. Poboiko, I. V. Gornyi, and A. D. Mirlin, “Measurement-Induced Phase Transition for Free Fermions above One Dimension,” *Phys. Rev. Lett.* **132**, 110403 (2024). [BPJ] [ARXIV]

Summary. One-loop renormalization of an $SU(R)$ replica sigma model predicts a genuine localization-type MIPT for free fermions in spatial dimension greater than one. Square-lattice numerics locate the two-dimensional transition and estimate its correlation-length exponent, supplying the central higher-dimensional benchmark.

- 32.** K. Chahine and M. Buchhold, “Entanglement Phases, Localization, and Multifractality of Monitored Free Fermions in Two Dimensions,” *Phys. Rev. B* **110**, 054313 (2024). [BPJ] [ARXIV]

Summary. An exact replica master equation, an AIII nonlinear sigma model, and Gaussian trajectory numerics characterize the two-dimensional monitored-fermion transition. The weak-monitoring phase has Fermi-liquid-like $L \log L$ entanglement, while the transition displays strong multifractality and emergent conformal signatures.

- 33.** I. Poboiko, I. V. Gornyi, and A. D. Mirlin, “Mesoscopic Fluctuations and Multifractality at and across Measurement-Induced Phase Transitions,” *Phys. Rev. B* **112**, L220202 (2025). [NEW] [ARXIV]

Summary. Two-dimensional trajectory numerics establish monitored analogues of universal conductance fluctuations, log-normal localized statistics, and scale-invariant distributions at criticality. The density correlator exhibits a measured multifractal spectrum, extending the Anderson analogy from averages to full

trajectory distributions.

- 34.** I. Poboiko, M. Szyniszewski, C. J. Turner, I. V. Gornyi, A. D. Mirlin, and A. Pal, “Measurement-Induced Lévy Flights of Quantum Information,” *Phys. Rev. Lett.* **135**, 170403 (2025). [NEW] [ARXIV]

Summary. Theory and large-scale numerics show that noncommuting measurements of misaligned, nonorthogonal local orbitals can null a quasiparticle decay rate and generate Levy-flight information transport with entanglement scaling as length to the one-third power. Away from maximal frustration, the system crosses successively through superdiffusive, logarithmic, and localized regimes.

- 35.** K. Chahine and M. Buchhold, “Spectral Transitions of the Entanglement Hamiltonian in Monitored Free Fermions,” *Phys. Rev. B* **112**, 214309 (2025). [NEW] [ARXIV]

Summary. Extensive numerics use entanglement-Hamiltonian eigenvalues and eigenvectors to distinguish chaotic, Fermi-liquid, and quantum-Lifshitz fixed points. Short- and long-range spectral statistics sharply diagnose localization, while one participation diagnostic suggests a possible intermediate multifractal regime.

- 36.** T. M. Mueller, M. Buchhold, and S. Diehl, “Monitored Interacting Dirac Fermions,” *Phys. Rev. B* **111**, 174201 (2025). [NEW] [ARXIV]

Summary. Bosonized replica field theory predicts an interaction-enabled BKT transition between algebraic and localized monitored Dirac phases. In the free limit the critical monitoring strength collapses to zero, reinforcing that short-range one-dimensional free fermions possess only a fine-tuned critical endpoint rather than a finite critical phase.

- 37.** B. Fan and A. M. García-García, “Entanglement Transitions and Multifractality in Monitored Free Fermions with Random Long-Range Hopping,” arXiv:2607.23871 (2026), preprint. [NEW] [ARXIV]

Summary. Large-scale trajectory numerics find volume or subvolume entanglement for sufficiently long-ranged random hopping and an alpha-dependent transition to an area law. At the transition, logarithmic entanglement, scale-invariant mutual information, and multifractal density correlations distinguish the long-range problem from short-range one-dimensional chains.

- 38.** C. Yin, B. Fan, and A. M. García-García, “No Measurement-Induced Phase Transition in the Entanglement Dynamics of Monitored Noninteracting One-Dimensional Fermions in a Disordered or Quasiperiodic Potential,” arXiv:2605.10758 (2026), preprint. [NEW] [ARXIV]

Summary. GPU trajectory simulations up to length 18000 find area-law entanglement at every nonzero monitoring strength for disordered and quasiperiodic one-dimensional chains. An NLSM with a disorder-induced mass supports the conclusion and attributes earlier apparent transitions to correlation lengths comparable with simulated sizes.

- 39.** I. Poboiko and A. D. Mirlin, “Quantum Dynamics of Monitored Free Fermions: Evolution of Quantum Correlations and Scaling at Measurement-Induced Phase Transitions,” *Phys. Rev. B* **113**, 144311 (2026). [NEW] [ARXIV]

Summary. Finite-time nonlinear sigma-model boundary conditions track how correlations grow from different initial states toward the monitored steady ensemble, with numerical confirmation. In the quasi-one-dimensional long-time regime the same localization time controls purification and charge sharpening, giving direct scaling predictions for preparation times near the two-dimensional MIPT.

V. TOPOLOGY, LYAPUNOV SPECTRA, AND DYNAMICAL BULK-BOUNDARY CORRESPONDENCE

How topology is encoded in nonunitary evolution, steady ensembles, and protected boundary sectors.

- 40.** A. Altland and M. R. Zirnbauer, “Nonstandard Symmetry Classes in Mesoscopic Normal-Superconducting Hybrid Structures,” *Phys. Rev. B* **55**, 1142–1161 (1997). [BOTH] [DOI]

Summary. Symmetry analysis introduced the superconducting random-matrix classes C, CI, D, and DIII and placed them on Cartan symmetric spaces. Together with the Wigner-Dyson and chiral classes, this is the equilibrium classification inherited by monitored Gaussian transfer matrices.

- 41.** S. Ryu, A. P. Schnyder, A. Furusaki, and A. W. W. Ludwig, “Topological Insulators and Superconductors: Tenfold Way and Dimensional Hierarchy,” *New J. Phys.* **12**, 065010 (2010). [BOTH] [ARXIV]

Summary. This classification constructs representatives and invariants for topological insulators and superconductors in every Altland-Zirnbauer class and dimension. Its dimensional hierarchy is the equilibrium table reused for spacetime topology in monitored

V. TOPOLOGY AND MONITORED DYNAMICAL PHASES—ADDITIONS

41a. A. Lavasani, Y. Alavirad, and M. Barkeshli, “Measurement-Induced Topological Entanglement Transitions in Symmetric Random Quantum Circuits,” *Nature Physics* **17**, 342–347 (2021). [NEW] [[arXiv](#)] [[DOI](#)]

Summary. Competing measurements and symmetry-preserving random unitaries produce distinct symmetry-protected topological and trivial area-law phases, together with a volume-law phase. In the measurement-only limit, the transition maps to two copies of two-dimensional percolation; sparse unitaries split this tricritical point into two critical lines. The work is a foundational example in which topology belongs to the ensemble of monitored trajectories and is visible in trajectory entanglement rather than ordinary linear expectation values.

41b. A. Lavasani, Y. Alavirad, and M. Barkeshli, “Topological Order and Criticality in $(2 + 1)$ D Monitored Random Quantum Circuits,” *Phys. Rev. Lett.* **127**, 235701 (2021). [NEW] [[arXiv](#)] [[DOI](#)]

Summary. Random Clifford unitaries, single-qubit measurements, and toric-code stabilizer measurements generate topologically ordered, trivial area-law, volume-law, and critical trajectory phases in two spatial dimensions. The phase diagram contains a percolation tricritical point and a distinct anisotropic self-dual tricritical point. This is the direct higher-dimensional monitored-circuit precedent for asking how a spatially patterned dynamical phase terminates at an interface.

- 51.** T. Iadecola, S. Ganeshan, J. H. Pixley, and J. H. Wilson, “Measurement and Feedback Driven Entanglement Transition in the Probabilistic Control of Chaos,” *Phys. Rev. Lett.* **131**, 060403 (2023). [BPJ] [ARXIV]

Summary. Numerics exhibit a diffusive transition between chaotic volume-law dynamics and a measurement-feedback-controlled disentangled phase, accompanied by a local order parameter. It establishes that feedback can make an entanglement transition visible in ordinary observables without trajectory post-selection.

- 52.** V. Ravindranath, Y. Han, Z.-C. Yang, and X. Chen, “Entanglement Steering in Adaptive Circuits with Feedback,” *Phys. Rev. B* **108**, L041103 (2023). [BPJ] [ARXIV]

Summary. This numerical and analytical circuit study finds separate trajectory-entanglement and averaged-density-matrix steering transitions. An explicit mapping of the latter to a parity-conserving branching-annihilating random walk shows how feedback symmetry changes the absorbing-state universality class.

- 53.** D. Alcalde Puente, F. Motzoi, T. Calarco, G. Morigi, and M. Rizzi, “Quantum state preparation via engineered ancilla resetting,” *Quantum* **8**, 1299 (2024). [BPJ] [ARXIV]

Summary. Matrix-product-state and trajectory numerics show that periodically resetting an ancilla can prepare the AKLT state and that a finite reset interval can outperform the infinitesimal Lindblad limit. The result is especially relevant to adaptive fermion protocols because it isolates when stroboscopic system-ancilla entanglement is a resource rather than an error.

- 54.** H. Pan, S. Ganeshan, T. Iadecola, J. H. Wilson, and J. H. Pixley, “Local and nonlocal stochastic control of quantum chaos: Measurement- and control-induced criticality,” *Phys. Rev. B* **110**, 054308 (2024). [NEW] [ARXIV]

Summary. Theory and numerics show that global control can lock measurement and control transitions together whereas local control generically separates them. This is a useful control for claims that activity or criticality is topological rather than a consequence of nonlocal feedback architecture.

- 55.** K. C. Smith, A. Khan, B. K. Clark, S. M. Girvin, and T.-C. Wei, “Constant-depth preparation of matrix product states with adaptive quantum circuits,” *PRX Quantum* **5**, 030344 (2024). [BPJ] [ARXIV]

Summary. The authors give explicit exact constant-depth adaptive preparations for a broad class of matrix-product states, including SPT and symmetry-broken examples. It is a sharp benchmark for the power gained from mid-circuit measurement and feedforward beyond finite-depth unitary circuits.

- 56.** N. O’Dea, A. Morningstar, S. Gopalakrishnan, and V. Khemani, “Entanglement and Absorbing-State Transitions in Interactive Quantum Dynamics,” *Phys. Rev. B* **109**, L020304 (2024). [BPJ] [ARXIV]

Summary. An exact classical mapping and quantum-trajectory numerics demonstrate that the absorbing transition of the averaged channel and the entanglement transition of trajectories are generically distinct. Product and cluster-state targets further show that target correlations and feedback constraints control both convergence and universality.

- 57.** A. Pocklington and A. A. Clerk, “Accelerating Dissipative State Preparation with Adaptive Open Quantum Dynamics,” *Phys. Rev. Lett.* **134**, 050603 (2025). [BPJ] [ARXIV]

Summary. This theoretical work shows that minimal adaptive open-system dynamics can evade a generic time-entanglement tradeoff and stabilize maximally entangled states on a finite timescale. Its fermionic motivation and parallel discrete/continuous constructions make it an immediate comparison point for adaptive Gaussian preparation.

- 58.** T. Doerstel, T. Iadecola, J. H. Wilson, and M. Buchhold, “Frustration-Free Control and Absorbing-State Transport in Entangled State Preparation,” *Phys. Rev. Lett.* **136**, 120603 (2026). [NEW] [ARXIV]

Summary. The paper introduces frustration-free control as a monitored preparation framework whose target is a common dark state of all local controllers. A solvable absorbing-walk mapping and Motzkin/Fredkin numerics show that nonlocal conserved charges make preparation transport limited, providing the closest established language to proposed topological frustration.

VII. DISSIPATIVE AND LINDBLADIAN PREPARATION, TOPOLOGY, AND CONTINUOUS ERROR CORRECTION

What local open dynamics can prepare, how quickly it mixes, and where topology imposes obstructions.

- 59.** J. P. Paz and W. H. Zurek, “Continuous Error Correction,” *Proc. R. Soc. London A* **454**, 355–364 (1998). [BPJ] [ARXIV]

Summary. This foundational proposal embeds stabilizer-code correction operations, coherent dynamics, and environmental errors in a continuous-time Markovian master equation. It provides historical context for using a Lindbladian as a controlled embedding of repeated correction channels.

- 60.** C. Ahn, H. M. Wiseman, and G. J. Milburn, “Quantum error correction for continuously detected errors,” *Phys. Rev. A* **67**, 052310 (2003). [BPJ] [ARXIV]
Summary. The authors derive an explicit feedback-control scheme that exactly protects a stabilizer codespace when each physical-qubit error channel is continuously and perfectly detected. The jump and diffusive versions make clear that physical continuous correction requires a classical measurement record and conditional control, not only a non-Hermitian no-click evolution.
- 61.** S. Diehl, E. Rico, M. A. Baranov, and P. Zoller, “Topology by Dissipation in Atomic Quantum Wires,” *Nature Phys.* **7**, 971–977 (2011). [BPJ] [ARXIV]
Summary. This proposal constructs a Lindblad reservoir that drives a one-dimensional atomic fermion wire into a paired topological steady state with Majorana edge modes. It established the dissipative gap and decoherence-free edge subspace as open-system counterparts of Hamiltonian bulk and boundary structure.
- 62.** C.-E. Bardyn, M. A. Baranov, C. V. Kraus, E. Rico, A. İmamoğlu, P. Zoller, and S. Diehl, “Topology by dissipation,” *New J. Phys.* **15**, 085001 (2013). [BPJ] [ARXIV]
Summary. The authors develop a density-matrix-based topological classification for quasi-local fermionic dissipative dynamics and analyze Majorana boundary modes. The theory also emphasizes that the damping spectrum and steady-state topology are independent structures, a distinction essential in interpreting Lindbladian edge modes.
- 63.** A. Lucia, T. S. Cubitt, S. Michalakis, and D. Pérez-García, “Rapid mixing and stability of quantum dissipative systems,” *Phys. Rev. A* **91**, 040302(R) (2015). [NEW] [ARXIV]
Summary. This theorem proves stability of local observables and correlations under local Liouvillian perturbations when the dynamics has a unique fixed point and mixes rapidly. It explains why a preparation proposal needs a mixing or convergence statement, not merely a formal dark state and spectral gap.
- 64.** J. C. Budich, P. Zoller, and S. Diehl, “Dissipative preparation of Chern insulators,” *Phys. Rev. A* **91**, 042117 (2015). [BPJ] [ARXIV]
Summary. This theoretical proposal constructs short-range system-bath couplings that generate a Chern-insulator steady-state phase and supplies a cold-atom implementation. It is a key antecedent whose purity, locality, and dissipative-gap tradeoffs are sharpened by later no-go results.
- 65.** M. Goldstein, “Dissipation-induced topological insulators: A no-go theorem and a recipe,” *SciPost Phys.* **7**, 067 (2019). [BPJ] [ARXIV]
Summary. Goldstein proves that above one dimension a finite-range quadratic Lindbladian cannot approach a unique pure Gaussian topological state at a nonvanishing exponential rate. The accompanying recipes show how exponentially local dynamics, or finite-range preparation of an arbitrarily close mixed state, evade the theorem.
- 66.** S. Lieu, M. McGinley, and N. R. Cooper, “Tenfold Way for Quadratic Lindbladians,” *Phys. Rev. Lett.* **124**, 040401 (2020). [NEW] [ARXIV]
Summary. The paper classifies the effective single-particle matrices of quadratic Lindbladians into ten Bernard-LeClair symmetry classes and relates their topology to finite-lifetime boundary excitations. Crucially, these spectral edge modes need not encode the topology of the steady state, preventing a common category error in Lindbladian topology.
- 67.** A. Altland, M. Fleischhauer, and S. Diehl, “Symmetry Classes of Open Fermionic Quantum Matter,” *Phys. Rev. X* **11**, 021037 (2021). [NEW] [ARXIV]
Summary. Starting from Fock-space transformations, the authors classify reversible, dissipative, and noise generators for interacting open fermionic matter. The framework separates nonequilibrium anti-linear symmetries from equilibrium time reversal and illustrates dissipative stabilization of an interacting SPT phase.
- 68.** F. Yang, P. Mognini, and E. J. Bergholtz, “Dissipative Boundary State Preparation,” *Phys. Rev. Research* **5**, 043229 (2023). [BPJ] [ARXIV]
Summary. The paper gives a loss-engineering recipe that makes selected SSH or Chern boundary states uniquely long lived while damping the remaining spectrum. For mirror-symmetric models it derives the complete Liouvillian spectrum and dissipative gap, offering a concrete spectral benchmark for interface-selective preparation.
- 69.** L. Mao, F. Yang, and H. Zhai, “Symmetry-preserving quadratic Lindbladian and dissipation driven topological transitions in Gaussian states,” *Rep. Prog. Phys.* **87**, 070501 (2024). [NEW] [ARXIV]
Summary. The authors derive conditions under which quadratic Lindblad evolution preserves the symmetry class of a Gaussian state’s modular Hamiltonian. Finite-size numerics then show finite-time dissipation-driven topological transitions in AIII and DIII examples, explicitly focusing on state topology rather than only Liouvillian spectrum topology.

70. Z. Ding, C.-F. Chen, and L. Lin, “Single-ancilla ground state preparation via Lindbladians,” *Phys. Rev. Research* **6**, 033147 (2024). [BPJ] [ARXIV]

Summary. The authors design continuous- and discrete-time dissipative ground-state algorithms implementable with one ancilla qubit, including cases with zero initial ground-state overlap. Ising- and Hubbard-model simulations establish relevance to fermionic quantum algorithms while clearly treating the Lindbladian as an engineered algorithm rather than a microscopic bath approximation.

71. J. Guo, O. Hart, C.-F. Chen, A. J. Friedman, and A. Lucas, “Designing open quantum systems with known steady states: Davies generators and beyond,” *Quantum* **9**, 1612 (2025). [BPJ] [ARXIV]

Summary. This theoretical framework characterizes nearly all Hamiltonian and dissipative terms compatible with a prescribed stationary state and generalizes Davies construction beyond equilibrium. It also derives syndrome-measurement and local-feedback corrections for incompatible noise, linking engineered steady states directly to autonomous error correction.

72. L. Mao and F. Yang, “Symmetry classification correspondence between quadratic Lindbladians and their steady states,” *Phys. Rev. B* **112**, 085136 (2025). [NEW] [ARXIV]

Summary. This classification analyzes generic particle-conserving quadratic Lindbladians in 27 non-Hermitian symmetry classes and identifies the five classes that correspond one-to-one with nontrivial Hermitian steady-state classes. Random-Lindbladian numerics verify the predicted long-time symmetry correspondence and expose how often the steady state is instead infinite temperature.

73. A. Chaduteau, D. K. K. Lee, and F. Schindler, “Lindbladian versus Postselected non-Hermitian Topology,” *Phys. Rev. Lett.* **136**, 016603 (2026). [NEW] [ARXIV]

Summary. The paper proves precise relations between the winding of a quadratic Lindbladian and that of its postselected non-Hermitian Hamiltonian in pure-loss or pure-gain limits. With simultaneous gain and loss it finds full-Lindbladian transitions and skin-effect reversals invisible to postselection, sharply demonstrating why quantum jumps cannot generally be discarded.

74. Y. Zhan, Z. Ding, J. Huhn, J. Gray, J. Preskill, G. K.-L. Chan, and L. Lin, “Rapid Quantum Ground State Preparation via Dissipative Dynamics,” *Phys. Rev. X* **16**, 011004 (2026). [BPJ] [ARXIV]

Summary. The paper proves sharp polynomial mixing bounds for quasi-free cases and rapid mixing for classes of weakly interacting spin and fermion systems, alongside tensor-network numerics for noncommuting

Hamiltonians. It supplies modern convergence tools for distinguishing a small spectral gap from the actual trace-distance preparation time.

VIII. CHERN-STATE PREPARATION, WANNIER OBSTRUCTIONS, AND REAL-SPACE DIAGNOSTICS

The single-particle geometry needed to formulate local controllers and verify their targets.

75. C. Brouder, G. Panati, M. Calandra, C. Mourougane, and N. Marzari, “Exponential localization of Wannier functions in insulators,” *Phys. Rev. Lett.* **98**, 046402 (2007). [BPJ] [ARXIV]

Summary. This theorem identifies exponentially localized Wannier bases with analytic periodic Bloch frames and proves their existence for time-reversal-symmetric insulators. Its converse shows that nonzero Chern number obstructs a complete exponentially localized Wannier basis.

76. R. Bianco and R. Resta, “Mapping topological order in coordinate space,” *Phys. Rev. B* **84**, 241106(R) (2011). [BPJ] [ARXIV]

Summary. This paper derives the local Chern marker, a projector-based coordinate-space diagnostic whose bulk average recovers the Chern number and whose spatial variation resolves interfaces. Model numerics validate it under open and periodic settings, making it the natural real-space monitor for prepared Chern domains.

77. J. Dubail and N. Read, “Tensor network trial states for chiral topological phases in two dimensions and a no-go theorem in any dimension,” *Phys. Rev. B* **92**, 205307 (2015). [FGTN] [ARXIV]

Summary. The paper constructs Gaussian tensor-network trial states for chiral free-fermion phases and proves that any strictly short-range quadratic parent Hamiltonian in the stated class must be gapless. The same algebraic argument rules out compactly supported Wannier-type generators, making it central to any finite-range topological preparation claim.

78. S. Barbarino, J. Yu, P. Zoller, and J. C. Budich, “Preparing Atomic Topological Quantum Matter by Adiabatic Nonunitary Dynamics,” *Phys. Rev. Lett.* **124**, 010401 (2020). [BPJ] [ARXIV]

Summary. This proposal prepares a Chern insulator by coupling it to a conjugate system, following an adiabatic path in the enlarged system, and discarding the conjugate copy. It is a close nonunitary predecessor to ancilla-assisted adaptive preparation and generalizes to invertible topological phases.

79. R. Fan, P. Zhang, and Y. Gu, “Generalized Real-space Chern Number Formula and Entanglement Hamiltonian,” *SciPost Phys.* **15**, 249 (2023). [BPJ] [ARXIV]

Summary. The authors generalize real-space Chern formulas to higher order and prove, for free fermions, proposed relations between the entanglement Hamiltonian and electric or thermal Hall response. The work supplies projector-level diagnostics that can be evaluated on Gaussian trajectory states without translational symmetry.

80. N. Baù and A. Marrazzo, “Local Chern marker for periodic systems,” *Phys. Rev. B* **109**, 014206 (2024). [NEW] [ARXIV]

Summary. The authors derive a local Chern marker compatible with large periodic supercells, avoiding the ill-defined ordinary position operator under periodic boundary conditions. Tight-binding tests on clean, disordered, and topological/trivial superlattices make it directly useful for finite-size domain-wall simulations.

81. Q. Li, J. Dong, P. J. Ledwith, and E. Khalaf, “Constraints on real space representations of Chern bands,” arXiv:2407.02561 (2024), preprint. [BPJ] [ARXIV]

Summary. The authors prove complementary real-space descriptions of a Chern band: complete orthogonal functions with universal power-law tails, or exponentially localized nonorthogonal overcomplete coherent states. They also derive a saturable spread bound, directly justifying the overcomplete localized frames used in adaptive Chern preparation.

82. T. Cole and D. Vanderbilt, “Reduced Wannier representation for topological bands,” *Phys. Rev. B* **111**, 205139 (2025). [NEW] [ARXIV]

Summary. This construction extracts exponentially localized Wannier functions for a proper subspace of topologically obstructed bands by enlarging the unit cell. It cleanly separates a localized trivial sector from an itinerant sector that retains the topology, offering a useful control for subspace-based counting and preparation.

IX. FERMIONIC QUANTUM SIMULATORS, PROCESSORS, AND EXPERIMENTAL PLATFORMS

Architectures capable of implementing fermionic modes, measurements, ancillas, and feedback.

83. L. W. Cheuk *et al.*, “A Quantum Gas Microscope for Fermionic Atoms,” *Phys. Rev. Lett.* **114**, 193001 (2015). [NEW] [ARXIV]

Summary. This experiment realized single-site-resolved imaging of fermionic potassium with detection fidelity above 95 percent while keeping atoms near the motional ground state. It supplies the local number readout, defect diagnosis, and low-entropy assembly capabilities that make mode-resolved monitored protocols experimentally meaningful.

84. D. González-Cuadra *et al.*, “Fermionic quantum processing with programmable neutral atom arrays,” *Proc. Natl. Acad. Sci. USA* **120**, e2304294120 (2023). [BPJ] [ARXIV]

Summary. This proposal encodes fermionic modes directly in movable neutral atoms and constructs non-local tunneling and interaction gates without paying a fermion-to-qubit string overhead. It is the closest hardware-level route to implementing adaptive mode measurements and conditional fermionic transport with statistics enforced by the register itself.

85. E. Crane *et al.*, “Hybrid Oscillator–Qubit Quantum Processors: Simulating Fermions, Bosons, and Gauge Fields,” arXiv:2409.03747 (2024), preprint. [BPJ] [ARXIV]

Summary. This proposal develops a hybrid oscillator-qubit simulation framework with exact and approximate decompositions for particle interactions, nonlocal observables, ground-state estimation, and partial error detection. It broadens the implementation landscape beyond native fermionic atoms and identifies ancilla-rich hardware primitives that can host feedback and monitored-particle protocols.

86. A. Schuckert, E. Crane, A. V. Gorshkov, M. Hafezi, and M. J. Gullans, “Fault-tolerant fermionic quantum computing,” arXiv:2411.08955 (2025), preprint. [BPJ] [ARXIV]

Summary. This proposal constructs fermionic repetition and color codes together with a native logical gate set, including qubit-controlled fermionic evolution. It clarifies how parity, loss, and phase errors should be represented when monitored fermionic dynamics is elevated from a state-preparation primitive to a fault-tolerant processor operation.

87. C. Tabares, C. Kokail, P. Zoller, D. González-Cuadra, and A. González-Tudela, “Programming Optical-Lattice Fermi-Hubbard Quantum Simulators,” *PRX Quantum* **6**, 030356 (2025). [NEW] [ARXIV]

Summary. This proposal compiles native Fermi-Hubbard controls into faster variational-adiabatic circuits and a variational approximation to imaginary-time evolution. It directly connects experimentally programmable fermionic gates to ground-state prepa-

ration and shows where nonunitary primitives can be emulated without treating them as literal deterministic evolution.

- 88.** R. Ott *et al.*, “Error-Corrected Fermionic Quantum Processors with Neutral Atoms,” *Phys. Rev. Lett.* **135**, 090601 (2025). [BPJ] [ARXIV]

Summary. This proposal introduces a finite fermionic reference that circumvents atom-number superselection and enables logical fermionic modes and number-conserving gates. The explicit reference and ancilla bookkeeping are directly relevant when a preparation controller must remain globally number conserving even though local correction steps exchange particles.

- 89.** G. Calliari *et al.*, “Programmable Fermionic Quantum Processors with Globally Controlled Lattices,” arXiv:2604.13160 (2026), preprint. [NEW] [ARXIV]

Summary. This proposal proves universality and supplies constructive protocols for arbitrary fermionic processes using global tunneling, gradient, and interaction controls in optical lattices. Its globally controlled architecture is especially pertinent to scaling repeated local-mode controllers without assuming fully independent addressability at every step.

- 90.** P. Bojović *et al.*, “High-fidelity collisional quantum gates with fermionic atoms,” *Nature* **652**, 602–608 (2026). [NEW] [ARXIV]

Summary. This experiment demonstrates spin-exchange and pair-tunneling entangling gates with fidelities up to 99.75 percent and long-lived Bell states in a fermionic optical superlattice. It turns the native-gate proposals into an experimentally grounded route for controlled particle transfers, local readout, and complex-state preparation.

X. CONTINUOUS MONITORING, QUANTUM TRAJECTORIES, COUNTING, AND FCS BRIDGES

The distinction between physical records, tilted generating functions, and detector backaction.

- 91.** L. S. Levitov, H.-W. Lee, and G. B. Lesovik, “Electron Counting Statistics and Coherent States of Electric Current,” *J. Math. Phys.* **37**, 4845–4866 (1996). [NEW] [ARXIV]

Summary. This foundational theory derives the determinant generating function for charge transferred through a coherent conductor using an explicit counting-field detector construction. It defines counting as statistics of a time-integrated transfer record, not as continuous projective measurement of an instantaneous fermionic occupation.

- 92.** I. Klich, “Full Counting Statistics: An elementary derivation of Levitov’s formula,” in *Quantum Noise in Mesoscopic Physics*, pp. 397–402 (Kluwer, 2003). [NEW] [ARXIV]

Summary. This derivation reduces Fock-space traces of fermionic bilinears to single-particle determinants and thereby makes coherent FCS computationally tractable. The identity is the natural Gaussian baseline against which record-generating measurements, dephasing, and feedback must be distinguished.

- 93.** J. P. Garrahan and I. Lesanovsky, “Thermodynamics of Quantum Jump Trajectories,” *Phys. Rev. Lett.* **104**, 160601 (2010). [NEW] [ARXIV]

Summary. This theory imports large-deviation thermodynamics into ensembles of counted quantum jumps and identifies dynamical phases through singularities of a tilted generator. It supplies the standard language for testing whether an interface supports persistent activity through its scaled cumulant-generating function rather than through average activity alone.

- 94.** J. E. Gough, M. I. Guță, M. R. James, and H. I. Nurdin, “Quantum filtering for systems driven by fermion fields,” *Commun. Inf. Syst.* **11**, 237–268 (2012). [NEW] [ARXIV]

Summary. This mathematical work derives filters for systems coupled to fermionic input fields and explains how even output observables such as number processes generate ordinary classical records. It directly addresses the parity-superselection concern: physical monitoring couples through parity-compatible system-field models even when odd fermionic operators appear in the microscopic interaction.

- 95.** N. T. Vidal, M. L. Bera, A. Riera, M. Lewenstein, and M. N. Bera, “Quantum Operations in an Information Theory for Fermions,” *Phys. Rev. A* **104**, 032411 (2021). [NEW] [ARXIV]

Summary. This work characterizes fermionic states and operations compatible with parity superselection and proves the equivalence of projective, Kraus, and dilation descriptions in that setting. It provides the clean operational test for whether a proposed counting channel is physical, independent of whether a convenient trajectory formula uses odd Kraus operators.

- 96.** G. T. Landi, M. J. Kewming, M. T. Mitchison, and P. P. Potts, “Current Fluctuations in Open Quantum Systems: Bridging the Gap Between Quantum Continuous Measurements and Full Counting Statistics,” *PRX Quantum* **5**, 020201 (2024). [NEW] [ARXIV]

Summary. This tutorial explicitly unifies stochastic output currents from jump or diffusive continuous measurements with tilted-generator full counting statistics. Its careful separation of instruments, currents, white

detector noise, cumulants, and waiting times is the most direct reference for turning a formal activity observable into a physical record statistic.

- 97.** E. Starchl, M. H. Fischer, and L. M. Sieberer, “Generalized Zeno Effect and Entanglement Dynamics Induced by Fermion Counting,” *PRX Quantum* **6**, 030302 (2025). [NEW] [ARXIV]

Summary. This paper studies continuously recorded fermion gain and loss while preserving total system-plus-environment number and conditional purity. It finds, analytically and numerically, a generalized Zeno regime whose correlations and entanglement resemble occupation monitoring even though the system state keeps fluctuating, making it the closest direct precedent for active fermion-counting dynamics.

- 98.** R. D. Soares, Y. Le Gal, and M. Schirò, “Entanglement Transition due to particle losses in a monitored fermionic chain,” *Phys. Rev. B* **111**, 064313 (2025). [NEW] [ARXIV]

Summary. This numerical study resolves local particle-loss trajectories in a paired free-fermion chain and computes waiting-time and entanglement-gain statistics. It demonstrates that jump records contain dynamical information absent from no-click evolution and that pairing can stabilize entangled trajectory steady states rather than an absorbing vacuum.

- 99.** F. Albarelli and M. G. Genoni, “Efficient evaluation of fundamental sensitivity limits and full counting statistics

for continuously monitored Gaussian quantum systems,” arXiv:2602.23304 (2026), preprint. [NEW] [ARXIV]

Summary. This proposal reduces two-sided and tilted generalized master equations for continuously monitored Gaussian systems to covariance and first-moment equations. Although formulated for bosonic linear systems, its unified treatment of jump and diffusive FCS offers a useful computational template for Gaussian trajectory diagnostics and explicitly identifies what changes when records are inefficient.

- 100.** F. Kloiber-Tollinger and L. M. Sieberer, “Replica Keldysh field theory of quantum-jump processes: General formalism and application to imbalanced and inefficient fermion counting,” *Phys. Rev. B* **114**, 024307 (2026). [NEW] [ARXIV]

Summary. This work derives a replica Keldysh theory for general jump processes, including mixed conditional states caused by undetected events, and applies it to imbalanced fermion gain and loss. It sharply separates efficient pure-state counting, inefficient monitoring, and the unconditional Lindblad limit, while predicting how missed records cut off long-distance entanglement.

Selection rule. Direct monitored-fermion, dynamical-topology, adaptive-preparation, and topological-Lindbladian papers are included broadly. Adjacent works are retained when their method is directly reused; generic background is represented only by canonical anchors. The companion ledger records the full seed census and screened new literature.